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$$\phi = E \cdot da$$

$$\phi = E\,da \cos\theta$$

$$\phi = E\,da \quad \text{if } \theta = 0^\circ$$

$$\phi = 0 \quad \text{if } \theta = 90^\circ$$

$$\phi = E\,da \cos 180^\circ \quad \text{if } \theta = 180^\circ$$

$$\phi = -E\,da$$

If area is not unit,

$$\phi_1 = E \cdot da_1$$

$$\phi_2 = E \cdot da_2$$

$$\vdots$$

$$\phi_n = E \cdot da_n$$

Total Flux

$$\phi_{total} = \phi_1 + \phi_2 + \dots + \phi_n$$

$$= E da_1 + E da_2 + \dots + E da_n$$

$$= E \cdot [da_1 + da_2 + \dots + da_n]$$

$$= E \cdot [\text{Sum of } da\text{'s}]$$

$$= E \cdot \int da$$

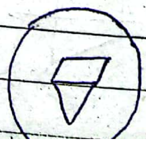
$$\phi_{total} = \int E \cdot da$$

Solid Angle

The angle substended by the two-dimensional to any point is known as Solid Angle.
Mathematically, it can be written as,

$$\text{Solid Angle} = \frac{\text{Vector Area}}{r^2}$$

$$d\Omega = \frac{da\ \hat{r}}{r^2}$$



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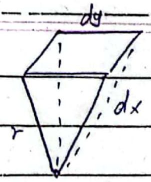


$$d\Omega = \frac{|da|\,\hat{n} \cdot \hat{r}}{r^2}$$

$$d\Omega = \frac{da\,|\hat{n}||\hat{r}|\cos\theta}{r^2}$$

$$\theta = 0^\circ$$

$$d\Omega = \frac{da\,|\hat{n}||\hat{r}|\cos 0^\circ}{r^2}$$

$$d\Omega = \frac{da}{r^2}$$

Integrating,

$$\int d\Omega = \frac{1}{r^2} \int da$$

$$\int d\Omega = \frac{1}{r^2} [a]$$

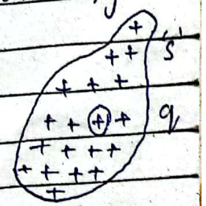
$$\int d\Omega = \frac{1}{r^2} [\text{Area of sphere}]$$

$$\int d\Omega = \frac{1}{r^2} [4\pi r^2]$$

$$\int d\Omega = 4\pi$$ Steradia is unit.

Flux due to N charges or Guass's law

consider N point charges $q_1, q_2, q_3, \dots, q_n$ enclosed by a surface 'S' as shown in figure. To determine electric flux due to these charges, through a closed surface we first determine the electric flux due to charge q_1



Date:

Suppose

$$\Phi = \int E \cdot da$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$\Phi = \int \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot da$$

$$= \frac{q}{4\pi\epsilon_0} \int \frac{da}{r^2} \hat{r}$$

$$= \frac{q}{4\pi\epsilon_0} \int \frac{|da|\, \hat{n} \cdot \hat{r}}{r^2}$$

$$\Phi_1 = \frac{q_1}{\cancel{4\pi}\epsilon_0} \times \cancel{4\pi}$$

$$= \frac{q_1}{\epsilon_0}$$

Similarly,

$$\Phi_2 = \frac{q_2}{\epsilon_0} \; ; \; \Phi_3 = \frac{q_3}{\epsilon_0} \; ; \; \Phi_n = \frac{q_n}{\epsilon_0}$$

Total flux:

$$\Phi_{total} = \Phi_1 + \Phi_2 + \Phi_3 + \dots + \Phi_n$$

$$= \frac{q_1}{\epsilon_0} + \frac{q_2}{\epsilon_0} + \frac{q_3}{\epsilon_0} + \dots + \frac{q_n}{\epsilon_0}$$

$$= \frac{1}{\epsilon_0} [q_1 + q_2 + q_3 + \dots + q_n]$$

$$= \frac{1}{\epsilon_0} [\text{Sum of } q\text{'s}]$$

Date:

$$= \frac{1}{\epsilon_0} \times Q_{total} \quad \text{: total charge}$$

The total flux due to N charges. It can be defined as the flux in E through any surface $= \frac{1}{\epsilon_0} \times q_{enclosed}$ described as integral form of Guass's law.

Flux due to N charge (differentiation)
{ Point (i) same & figure same }
We know that the integral form of Gauss's Law:

$$\Phi = \frac{q_{total}}{\varepsilon_0} \quad \text{— (i)}$$

$$\Phi = \int E \cdot da \quad \text{— (ii)}$$

From eq (i)

$$\Phi = \frac{q_v}{\varepsilon_0}$$

$$\Phi = \frac{\rho \int dv}{\varepsilon_0} \quad \because q = \rho \int dv$$

We know that

$$\Phi = \int E \cdot da$$

$$\int E \cdot da = \frac{\rho \int dv}{\varepsilon_0}$$

Transforming the surface integral into volume integral by Guass's Divergence Theorem.

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Guass's Divergence Theorem:

$$\int E \cdot da = \int div(E)\, dv$$

So the above eq can be written as:

$$\int div(E)\, dv = \frac{\rho}{\varepsilon_0}\int dv$$

$$div(E)\int dv = \frac{\rho}{\varepsilon_0}\int dv$$

$$div(E)\,[\cancel{V}] = \frac{\rho}{\varepsilon_0}\,[\cancel{V}]$$

$$div(E) = \frac{\rho}{\varepsilon_0}$$

or

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\because \nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\because E = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$$\nabla \cdot E = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(E_x\hat{i} + E_y\hat{j} + E_z\hat{k}\right) = \frac{\rho}{\varepsilon}$$

$$\frac{\partial}{\partial x}E_x + \frac{\partial}{\partial y}E_y + \frac{\partial}{\partial z}E_z = \frac{\rho}{\varepsilon_0}$$

This is ~~kno~~ the differentiation form of Guass's Law.

Fri Date: June 14

Q Humid air becomes ionized on E.F of 3.0×10^6 N/C

What is the electric force on an electron while the charge is 1.6×10^{-19} C

$q = 1.6\times10^{-19}$ C

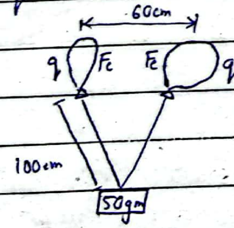
$E = 3.0\times10^6$ N/C

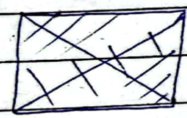
$F = ?$

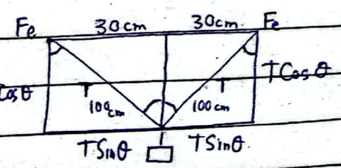
$F = qE = 1.6\times10^{-19}\times 3.0\times10^6$

$F = 4.8\times10^{-13}$ N Ans

Q. Two Similar He-filled balloons tied to a 50 gram weights, floats in equillibrium, total force on each balloon is zero. Each balloon has a charge "q" as shown in fig. Find the values of q.







Date:

$m = 50gm = 0.05\,kg$

$r = 60cm = 0.6m$

$\Sigma F_x = 0$

$2T\sin\theta - 2F_e = 0$

$T\sin\theta = F_e$

$\sin\theta = P/H = 30/100$

$\theta = 17.45°$

$W = mg$

$W = 0.05 \times 9.8$

$W = 0.49N$

$\Sigma F_y = 0$

$2T\cos\theta - W = 0$

$2T\cos\theta = W$

$T = \frac{W}{2\cos\theta}$

$\boxed{T = 0.256N}$

$F_e = T\sin\theta$

$F_e = 0.256\sin 17.45°$

$\boxed{F_e = 0.076N}$

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$F_e = \frac{kq^2}{r^2}$

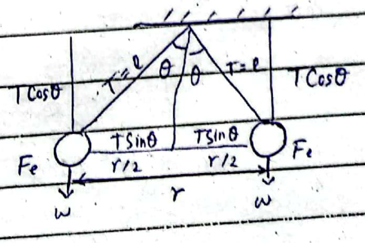
$0.076 = \frac{9\times10^9 \times q^2}{(0.6)^2}$

$q = 1.75\times10^{-6}C$

or $1.75\,\mu C$ Ans

Q Two small spheres of masses 'm' are suspended from common point of thread of length "l" where each sphere carries a charge "q" each thread makes an angle "θ" with the vertical as shown in fig. Show that charge "q" is given by:

$q = 2l\sin\theta\sqrt{\frac{mg\tan\theta}{k}}$



$\sin\theta = P/H$

$\sin\theta = \frac{r/2}{l}$

$\boxed{r = 2l\sin\theta}$

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Date:

$$\sum F_y = 0$$

$$2T\sin\theta = 2F_e$$

$$F_e = T\sin\theta$$

$$\frac{kq^2}{r^2} = T\sin\theta$$

$$T = \frac{kq^2}{r^2\sin\theta} \quad \text{—— (i)}$$

$$\sum F_y = 0$$

$$2T\cos\theta - 2W = 0$$

$$T\cos\theta = W$$

$$T = \frac{W}{\cos\theta} = \frac{mg}{\cos\theta} \quad \text{—— (ii)}$$

Equate (i) & (ii)

$$\frac{kq^2}{r^2\sin\theta} = \frac{mg}{\cos\theta}$$

$$kq^2 = \frac{mg\,r^2\sin\theta}{\cos\theta}$$

$$q = \sqrt{\frac{mg\,r^2\tan\theta}{k}}$$

$$q = r \times \sqrt{\frac{mg\tan\theta}{k}}$$

$$q = 2l\sin\theta\sqrt{\frac{mg\tan\theta}{k}}$$

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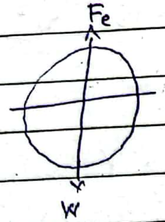
Q) A Water droplet 10^{-2} cm in diameter carrying a negative charge such that magnitude of Electric field at its surface is 60,000 V/m. How strong vertical field would be required to keep the droplet falling.

$E_1 = 60,000\ V/m$

$E_2 = ?$



$d = 10^{-2}\,cm = 10^{-4}\,m$

$r = 5 \times 10^{-5}\,cm$

Charge on Droplet:

$$\therefore E = \frac{kq}{r^2}$$

$$q = \frac{(60000)(5\times10^{-5})^2}{9\times10^9}$$

$$\boxed{q = 1.6\times10^{-4}\,C}$$

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$$\left\{ \begin{array}{l} E = F/q \\ W = m \end{array} \right.$$

$\therefore F_e = W$

$q \cdot E_2 = mg$

$E_2 = \frac{mg}{q}$

$\therefore e = \frac{m}{v}$

Volume of Droplet:

$\therefore V = 4/3 \, \pi r^3$

$V = \frac{4}{3} (3.14)(5 \times 10^{-5})^3$

$\boxed{V = 5.3 \times 10^{-13} \, m^3}$

Mass of Droplet:

$m = e \times V$

$m = 1 \times (5.3 \times 10^{-13})$

$m = 5.3 \times 10^{-13} \, kg$

① $\Rightarrow E_2 = \frac{(5.3 \times 10^{-13})(9.8)}{1.6 \times 10^{-14}}$

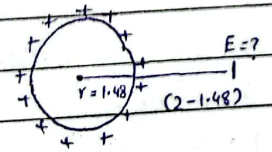
$E_2 = 324.7 \, V/m$

$E = E_1 + E_2$

$E = 60324.7 \, V/m$ Ans

Date: June 14

Q A uniformly charged spherical shell with radius 1.48 m at a surface charge density of $8.64 \, \mu C/m^2$. Find (i) Total charge on the shell ii) What is the total electric flux leaving the surface from the shell iii) Calculate the Electric field Intensity at the distance of 2 m from the centre of the shell.



$r = 1.48 \, m$

$\sigma = 8.64 \, \mu C/m^2$

i) $q = ?$

ii) $\phi = ??$

i) $\sigma = \frac{q}{A}$

$q = \sigma A$

$q = 8.64 \times 10^{-6} \times (4\pi r^2)$

$q = 8.64 \times 10^{-6} \times 4\pi \times (1.48)^2$

$q = 2.37 \times 10^{-4} \, C$

ii) Total E. flux

$\phi = \frac{q}{\epsilon_0} = \frac{2.37 \times 10^{-4}}{8.85 \times 10^{-12}}$

$\phi = 26.7 \times 10^{6}$ weber

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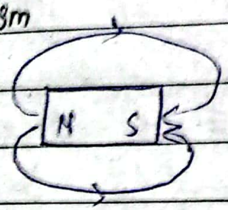
(ii) $E = \frac{kq'}{r^2} = \frac{9 \times 10^9 \times 2.37 \times 10^{-4}}{(2-1.48)^2}$

$E = 7.8 \times 10^8$ N/C

Magnetism



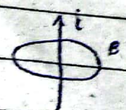
Assumptions:

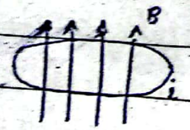
(i) Magnetic lines are always directed from North to South Pole.

(ii) Magnetic Field is represented by
a) arrows $\longrightarrow$
b) crosses × × ×.

(iii) Moving electron has a magnetic field. Moving proton has a magnetic field

(iv) Straight current-carrying conductor has circular mag. field

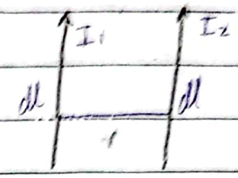




v) Circular current-carrying conductor has straight mag. field.

Date:

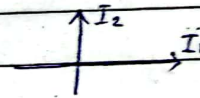
Force b/w two current carrying conductor



① Consider two conductors each of length dl carrying current I_1 & I_2 respectively and placed parallel to each other at a distance of r as shown in fig.

② Experiment shows that conductor attract each other if the current flows in the same direction. Conductors repel each other if the current flows in opposite direction.

③ If the conductors are placed in cross direction, no force will act.



④ Experiments shows that the force of attractions & repulsion b/w the conductors depend a) directly propotional to the length of the conductor

$F \propto dl$ — (i)

b) directly propotional to product of currents. $F \propto I_1 I_2$ — (ii)

c) Inversely propotional to the distance b/w the conductors

$$F \propto \frac{1}{r} \quad \text{— (iii)}$$

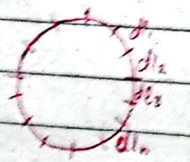
Combining (i), (ii) & (iii)

$$F \propto \frac{dl\, I_1 I_2}{r}$$

$$F = (\text{const}) \frac{dl\, I_1 I_2}{r}$$

$$F = \frac{\mu_0}{2\pi} \frac{dl\, I_1 I_2}{r}$$

This gives the force of attraction or repulsion b/w two current carrying conductor



$B = 2\pi r = \mu_0 I$

$B =$ orbital circumference $= \mu_0 I$

$= dl_1 + dl_2 + \dots + dl_n$

$=$ Sum of dl_s

$= \int dl$

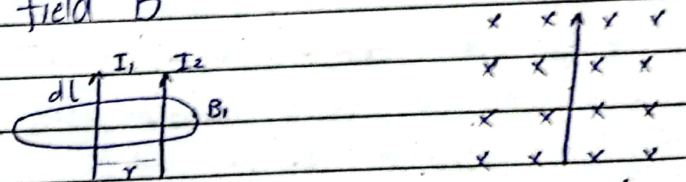
Ampere's Law (Integral form)

The line integral of the tangential magnetic field and the path element equal to μ_0 times the current enclosed by the circuit.

$$\boxed{\oint B \cdot dl = \mu_0 \times I}$$

1. Consider a conductor of length 'dl' carrying 'I' ampere of current surrounded by the magnetic field 'B'.

2. Let another current-carrying conductor of same length placed in the magnetic field B



$I\,dl\,B = F = I(dl \times B)$

3. The force exerted by the mag. field B on conductor 1 carrying I_2 ampere of current is given by.

$$F_0 = I_2 (dl \times B_1)$$

$$F = I_2\, dl\, B_1 \sin\theta \quad \because \theta = 90^\circ$$

$$F = I_2\, dl\, B_1 \quad \text{— (i)}$$

Date:

4 The force b/w two current-carrying conductor is also given by:

$$F = \frac{\mu_0}{2\pi} \frac{dl I_1 I_2}{r} \quad \text{—— (ii)}$$

Equate (i) & (ii)

$$I_2 dl B_1 = \frac{\mu_0}{2\pi} \frac{dl I_1 I_2}{r}$$

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$

or

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B \cdot 2\pi r = \mu_0 I$$

$$B \cdot (\text{orbital circumference}) = \mu_0 I$$

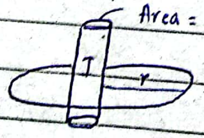
$$B \cdot (dl_1 + dl_2 + \dots + dl_n) = \mu_0 I$$

$$B \cdot (\text{Sum of } dls) = \mu_0 I$$

$$B \cdot \left(\int dl\right) = \mu_0 I$$

$$\int B \cdot dl = \mu_0 I \quad \text{proved!!}$$

Ampere's Law (Differential form)



Amperian loop

Date:

1 The integral form of Ampere's Law is written as

$$\int B \cdot dl = \mu_0 I \quad \text{—— (i)}$$

$$\text{current density} = \frac{\text{current}}{\text{Area}}$$

$$j = \frac{I}{dA}$$

$$I = j\,da \qquad dA = \int da$$

$$I = j \int da$$

Put the value of I in (i)

$$\int B \cdot dl = \mu_0 j \int da \quad \text{—— (ii)}$$

$\int B \cdot dl =$

Transforming line ~~&~~ integral to surface integral by using Stoke's Theorem

$$\int A \cdot dl = \int \text{curl } A \, da$$

$$\int B \cdot dl = \int \text{curl } B \, da$$

So above expression becomes

$$\int \text{curl } B \, da = \mu_0 j \int da$$

$$\text{curl } B \int da = \mu_0 j \int da$$

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$$\text{curl } B\,[a] = \mu_0 j\,[a]$$
$$\text{curl } B = \mu_0 j$$

This gives the differential form of Ampere's Law.
It shows that:
(i) B is produced by 'j' (current f
(ii) B is circular

$$\nabla \times B = \mu_0 j$$
$$\nabla\left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \times (B_x i + B_y j + B_z k) =$$

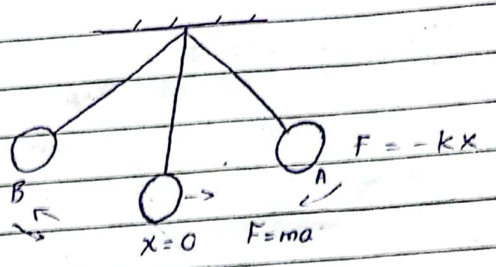
✓ Comparison b/w Electrostatic force & Magnetic force

i) Electrostatic force can be parallel or anti parallel	i) Mag. force is always perpendicular
ii) Independant of the velocity of charge	ii) dependant
iii) EF perform ~~two~~ work done on the charge q	iii) MF never perform workdone on q
iv) Conservative	iv) MF is not conservative

✓ Waves of Oscillation

Date: 21 - June

Simple Harmonic Motion:
To & fro motion of the body is called SHM.

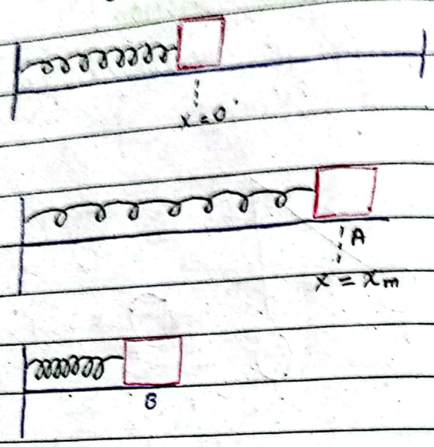


$$F = ma \quad \text{— (i)}$$
$$F = -kx \quad \text{— (ii)}$$

Equating (i) & (ii)
$$ma = -kx$$
$$a = -\frac{k}{m}x$$
$$a \propto -x$$

The Periodic Oscillation under linear restoring force is called SHM.
Examples of SHM
1. Mass Spring System
2. Simple Pendulum
3. Projection of the particle in verticle Circle

Mass Spring System



Newtonian Force

$$F = ma \quad \text{— (i)}$$

$$F = -kx \quad \text{— (ii)}$$

Equating (i) & (ii),

$$ma = -kx$$

$$a = -\frac{k}{m}x$$

$$a = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega^2 = \frac{k}{m}$$

$$a + \omega^2 x = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

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This is second order homogeneous differential equation which show that motion of the body is in SHM. The solution of this equation is given by

$$x(t) = x_m \cos(\omega t + \phi)$$

$$x(t) = x_m \sin(\omega t + \phi)$$

where $x(t)$ = displacement as a function of time; x_m is the amplitude of oscillation;
ω is angular velocity ($\omega^2 = k/m$)
ϕ = Phase angle / Constant
$(\omega t + \phi)$ is Phase of motion

Parameters of SHM:

1. Displacement:

$$\boxed{x(t) = x_m \cos(\omega t + \phi)} \text{ — (i)}$$

2. Velocity:

for velocity differentiate eq (i) w.r.t t

$$V(t) = \frac{dx}{dt} = \frac{d}{dt}\left[x_m \cos(\omega t + \phi)\right]$$

$$\boxed{V(t) = -x_m \omega \sin(\omega t + \phi)} \text{ (ii)}$$

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$V_m = \omega x_m$

So we can write,

$$\boxed{V(t) = -V_m \sin(\omega t + \phi)}$$

3. Acceleration:-

For acc diff. (ii) wrt t,

$$a(t) = \frac{dV(t)}{dt} = \frac{d}{dt}\left[-x_m \omega \sin(\omega t + \phi)\right]$$

$$a(t) = -x_m \omega^2 \cos(\omega t + \phi)$$

$$a_m = \omega^2 x_m$$

So,

$$\boxed{a(t) = -a_m \cos(\omega t + \phi)}$$

4. Time Period:-

$(\omega t + \phi)$

$(\omega(T+t) + \phi)$

$$(\omega(T+t) + \phi) - (\omega t + \phi) = 2\pi$$

$$\omega T + \omega t + \phi - \omega t - \phi = 2\pi$$

$$\omega T = 2\pi$$

$$\boxed{T = \frac{2\pi}{\omega}} = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

Date:

5. Frequency:

$$f = \frac{1}{T}$$

$$f = \frac{1}{2\pi/\omega}$$

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}}$$

Energy Consideration in SHM

Mechanical Energy remains constant but PE & K.E can fluctuate.

$$E = P.E + K.E$$

We know that,

$$PE = \frac{1}{2} kx^2 \quad \text{——— (i)}$$

We know

$$x = x_m \cos(\omega t + \phi)$$

Put the value of 'x' in (i)

$$P.E = \frac{1}{2} k \left[x_m \cos(\omega t + \phi)\right]^2$$

$$P.E = \frac{1}{2} k x_m^2 \cos^2(\omega t + \phi) \quad (A)$$

Date:

for K.E :

$$K.E = \frac{1}{2} m V^2 \quad — (ii)$$

$$V = -\omega x_m \sin(\omega t + \phi)$$

Put the value ~~in~~ of 'v' in (ii),

$$K.E = \frac{1}{2} m \left[-\omega x_m \sin(\omega t + \phi)\right]^2$$

$$K.E = \frac{1}{2} m \omega^2 x_m^2 \sin^2(\omega t + \phi)$$

$$K.E = \frac{1}{2} m . \frac{k}{m} . x_m^2 \sin^2(\omega t + \phi)$$

$$\boxed{K.E = \frac{1}{2} k x_m^2 \sin^2(\omega t + \phi)} \quad (B)$$

for Total Energy :

add (A) & (B)

$$E = K.E + P.E$$

$$E = \frac{1}{2} k x_m^2 \cos^2(\omega t + \phi) + \frac{1}{2} k x_m^2 \sin^2(\omega t + \phi)$$

$$\boxed{E = \frac{1}{2} k x_m^2}$$

Date:

at $x = -x_m$	at $x = 0$	at $x = x_m$
$PE = \frac{1}{2} k x_m^2$	$PE = 0$	$PE = \frac{1}{2} k x_m^2$
$KE = 0$	$K.E = \frac{1}{2} k x_m^2$	$K.E = 0$
$E = \frac{1}{2} k x_m^2$	$E = \frac{1}{2} k x_m^2$	$E = \frac{1}{2} k x_m^2$

Total Energy remains constant throughout the oscillation. In one cycle, we have two maximum values & two minimum values of Kinetic Energy.

Num 1 : The position of the particle is given by $x = 4\,m \cos(3\pi t + \pi)$ where x is in meters and t is in seconds. Calculate frequency, time period, phase constant, amplitude, max. velocity ($v_m = \omega x_m$), max. acceleration and position of the particle at $t = 0.25$ sec.

$$x(t) = x_m \cos(\omega t + \phi)$$

$$x = 4 \cos(3\pi t + \pi)$$

$$x_m = 4 \;, \quad \omega = 3\pi \;, \quad \phi = \pi$$

$$V(t) = -4 \times 3\pi \times \sin(3\pi t + \pi)$$

$$V(0.25) =$$

Date:

$a(t) = -4 \times (3\pi)^2 \times \cos(3\pi t + \pi)$

$a(0.25) =$

$f = \frac{\omega}{2\pi} = \frac{3\pi}{2\pi} = 3/2 \text{ Hz}$

$T = \frac{1}{f} = \frac{2}{3}$ sec

Num 2: A 1 kg object is attached to a spring with a force constant k equal to 25 N/m on a horizontal frictionless surface. At time t=0, the object ~~is at~~ is released from rest. If x = -3 cm, find the time period of motion, the max. speed, max. acc. and the function of position, velocity, acceleration wrt t

$m = 1$ kg $\quad k = 25$ N/m

at $t = 0 \quad x = -3$ cm

$\omega = \sqrt{k/m} = \sqrt{25/1} = 5$ rad/sec

$T = \frac{2\pi}{\omega} = \frac{2\pi}{5} = 1.256$ sec

Intro
Solution
Rubrics

Date:

$x = -3\text{cm} \Rightarrow -0.03\text{ m}$ $\quad V_m = \omega x_m$

So Amplitude $A = 0.03$m $\quad a_m = \omega^2 x_m$

$V_{max} = A\omega$ amplitude $\quad \therefore A = 3$ cm

$V_{max} = (3 \times 10^{-2})(5) = 15 \times 10^{-2}$ m/s

acc $A_{max} = A\omega^2 = (3 \times 10^{-2})(5)^2$ amp

$A_{max} = 75 \times 10^{-2}$ m/s

Position as a function of time:

$x(t) = A\cos(\omega t + \theta)$

Since the object is released from the rest $x = -3$cm at $t=0$ we need to determine ϕ.

$x(0) = A\cos(\phi) = -0.03$m

Since $A = 0.03$m

$0.03\cos(\phi) = -0.03$

$\cos(\phi) = -1$

Thus $\phi = \pi$

Now substituting the values

$x(t) = 0.03\cos(5t + \pi)$

{solve further by differentiating and using Time period's formula}

keypo

Date:

Q. A mass of 0.5kg connected to light string of force constant 20 N/m oscillates on a horizontal friction less surface.

(i) Calculate the total Energy of the system max velocity. Total Energy of man if the amplitude of motion is 3cm

(ii) What is the velocity of the man when displacement is equal to 2cm.

(iii) Find E_k and E_p when disp. is 2cm

(i) $E = \frac{1}{2} k x_0^2 = \frac{1}{2} k (0.03)^2$

$E = 9 \times 10^{-3}$ J

$V_{max} = \omega x_m = \sqrt{\frac{k}{k}}$

$V_{max} = \sqrt{\frac{k}{m}} \cdot x_m = \sqrt{\frac{20}{0.5}} \times 0.03$

$V_{max} = 0.1897$ m/s

(ii) $E = KE + PE$

$1/2\ k x_m^2 = 1/2\ m V^2 + 1/2 k x^2$

$20 \times (0.03)^2 = 0.5\ (V)^2 + \cancel{1/2}\ 20 \times (0.$

$V = 0.1414$ m/s

Date:

Q1 A force of 5N compress a spring by 5m

(i) Find k (ii) Find E_p of compressed spring.

Q2 A 100g object is suspended ~~by~~ with k 50 N/m.

(i) By how much does the spring stretch.

(ii) What is the period of oscillation

(iii) Find frequency

Q1. i) $F = kx$

$k = F/x = 5/5 = 1$ N/m

(ii) $PE = 1/2\ k x^2 = 1/2 (1)(5)^2 = 12.5$ J

Q2. (i) $W = kx = F$

$x = \frac{mg}{k} = \frac{0.1 \times 9.8}{50} = 0.01962$ m

(ii) $T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.1}{50}} = 0.281$ s

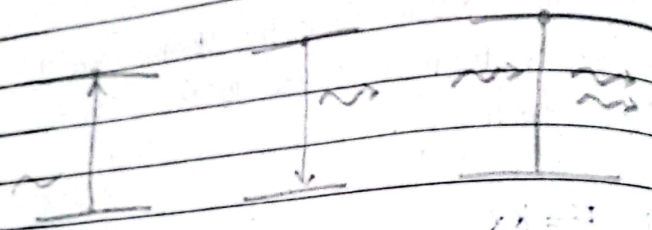
(iii) $f = 1/T = 3.56$ Hz

LASER

Date:

{LASER THEORY}

{Types of Emissions}



Requirement of Laser - action

Date:

Following condition must be satisfied to perform laser action

(i) Population Inversion
(ii) Optical Resonator
(iii) Lasing Medium
(iv) Means of excitation
v) Host medium

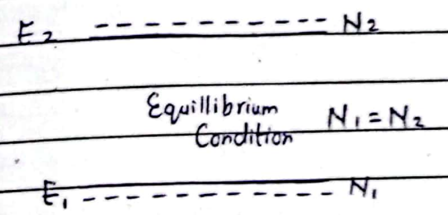


fig.

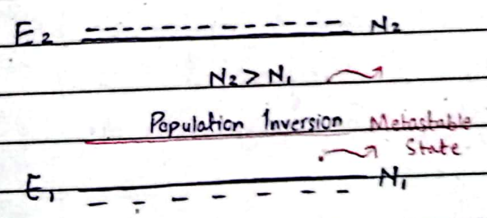
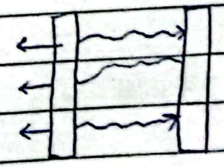


fig.

$$N_1 = N_0 e^{-E_1/KT}$$
$$N_2 = N_0 e^{-E_2/KT}$$



Date: ______

Q. Angle of Divergence :-

$$\phi = \frac{a_2 - a_1}{2(d_2 - d_1)}$$

→ Dia

→ Distance

Area of Spread / Divergence :-

$$A = (\phi \cdot D)^2$$

→ focal length

$$\text{Intensity} = \frac{P}{A}$$

Q. For a ruby laser, the output beam spot dias 1mm & 3mm at a distance of 3m & 6m. Calculate the angle of divergence & Area of Divergence 6m away from the lase

$$\phi = \frac{3 \times 10^{-3} - 1 \times 10^{-3}}{2(6-3)}$$

$$\phi = 3.33 \times 10^{-4} \text{ rad.}$$

$$A = (\phi \cdot D)^2$$

$$A = (3.33 \times 10^{-4} \times 6)^2$$

$$A = 4 \times 10^{-6} \text{ rad}^2 m^2$$

Date: ______

Q. A laser beam has a power of 100mW. It has aperture dia of 10mm and the wavelength of ~~100~~ 14400 Å. A beam is focal with a focal length of 0.1m. Calculate the Angle of Spread & Intensity of image

Angle of Spread :-

$$\phi = \frac{\lambda}{d}$$

d → aperture dia

$\lambda = 14400$ Å

$f = D = 0.1m$

$d = 10mm$

$$\phi = \frac{14400 \times 10^{-10}}{10 \times 10^{-3}} = 1.44 \times 10^{-4} \text{ rad}$$

for Intensity,

$$A = (\phi \cdot D)^2$$

$$A = (1.44 \times 10^{-4} \times 0.1)^2$$

$$A = 2.07 \times 10^{-10} \text{ rad}^2 m^2$$

$$I = P/A = \frac{100 \times 10^{-3}}{2.07 \times 10^{-10}}$$

$$I = 4.8 \times 10^8 \text{ W/m}^2$$

Date:

Q. A 3 level laser produce a light of 550 nm (λ). At what temperature, the ratio of population is half. $N_2/N_1 = 1/2$

$$\frac{N_2}{N_1} = \frac{N_0 e^{-E_2/KT}}{N_0 e^{-E_1/KT}} = \frac{1}{2}$$

$$e^{-E_2/KT + E_1/KT} = \frac{1}{2}$$

$$\frac{1}{2} = e^{-\Delta E/KT}$$

$\therefore \Delta E = h\nu$ Planks Law

taking ln both sides,

$$\ln 1 - \ln 2 = \ln e^{-\Delta E/KT}$$

$$-0.0693 = -\frac{h\nu}{KT} \times 1$$

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{550 \times 10^{-9}}$$

$$\nu = 5.4545 \times 10^{14}$$

$$T = \frac{h\nu}{K(0.0693)} = 37 \times 10^3 \, K$$

300K

$\frac{N_2}{N_1} = \frac{1}{2}$

Date:

Q At RTP, the ratio of $N_2/N_1 = 1/e$. Calculate the frequency of light produced by laser.

$$\frac{N_2}{N_1} = \frac{N_0 e^{-E_2/KT}}{N_0 e^{-E_1/KT}} = \frac{1}{e}$$

$$e^{-E_2/KT + E_1/KT} = \frac{1}{e}$$

$$e^{-\Delta E/KT} = \frac{1}{e}$$

$$\ln e^{-\Delta E/KT} = \ln e^{-1}$$

$$-\frac{\Delta E}{KT} = -1$$

$$+\frac{h\nu}{KT} = +1$$

$$\nu = \frac{KT}{h} = \frac{K(300)}{h}$$

$$\nu = 6.25 \times 10^{+12} \text{ Hz}$$

CLO-3 Date: 28 June

De-Broglie's Law

De-Broglie's Hypothesis
Photoelectric effect
Compton Effect
Black B. Radiation
Radioactivity
(Only Problems Included)

Light

Wave — Particle

Wave: light behaves as wave in propagated expt. e.g Interference Diffraction, polarization

Particle: (mentioned below)

→ Particle nature: Light behave as particle in interaction expt. e.g Photoelectric effect, Compton effect, Black body radiation.

→ Wave: (i) frequency (ii) Wavelength (iii) Amplitude (iv) Intensity

→ A particle is specified as
(i) Mass (ii) Velocity (iii) Momentum (iv) Energy

From Plank's Theory:

$$E = h\nu \quad \text{— (i)}$$

From Einstein Theory

$$E = mc^2 \quad \text{— (ii)}$$

Date: ______

Equating (i) & (ii),

$$mc^2 = h\nu$$

$$c(mc) = h\nu$$

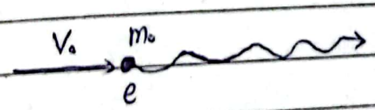
$$mc = \frac{h\nu}{c}$$

$$P = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{P}$$

$$\boxed{\lambda = \frac{h}{mv}}$$ for heavy particle

Application of De Broglie's Hypothesis:



$$KE = \frac{1}{2} m_0 v^2$$

$$KE = eV_0$$

$$\frac{1}{2} m_0 v^2 = eV_0$$

$$v^2 = \frac{2eV_0}{m_0}$$

$$v = \sqrt{\frac{2eV_0}{m_0}}$$

v = velocity $\quad \nu$ = freq

$\bar{\nu}$ = wave no.

Date:

A/C to De-Broglie's

$$\lambda = \frac{h}{m_0 v}$$

$$\lambda = \frac{h}{m_0 \sqrt{\frac{2eV_0}{m_0}}}$$

$$\lambda = \frac{h}{\sqrt{2eV_0 m_0}}$$

Q. An electron initially at rest accelerated through a potential difference of 54V. Compute (i) Velocity of the electron
(ii) De-Broglie's Wavelength
(iii) Wave no (Reciprocal of wavelength.)

(i) $$v = \sqrt{\frac{2eV_0}{m_0}}$$

$\therefore e =$

$\therefore m_0 =$

$$v = \sqrt{\frac{2e(54)}{m_0}} = 4.3 \times 10^8 \ m/s$$

Date:

(ii) $$\lambda = \frac{h}{\sqrt{2eV_0 m_0}} = \frac{h}{\sqrt{2e(54)m_0}}$$

$$\lambda = 1.67 \text{ Å}$$

(iii) $$\boxed{\bar{\nu} = \frac{1}{\lambda}}$$

Q. An electron initially at rest accelerated through a P.D of 5000V. Compute:
(i) momentum, (ii) De-Broglie's Wavelength
(iii) Wave no.

(i) $$P = m_0 v = m_0 \sqrt{\frac{2eV_0}{m_e}}$$

(ii) $$\lambda = \frac{h}{\sqrt{2e(5000)m_e}} = \frac{h}{P}$$

(iii) $$\bar{\nu} = \frac{1}{\lambda}$$

Q. Compute the De-Broglie's wavelength of a proton whose KE = rest energy of electron ~~energy~~. Mass of the proton is ~~1.836~~ 1836 times mass of electron

Date:

$$m_p = 1.836 \times m_0 \quad (1836)$$
$$m_p = 1.836 \times 9.1\times10^{-31}$$
$$m_p = 1.67\times10^{-27}\ kg$$

$$KE = mc^2$$
$$\frac{1}{2} m_p v^2 = m_0 c^2$$
$$v^2 = \frac{2m_0c^2}{m_p} = \frac{2m_0c^2}{1836\, m_0}$$
$$v = \sqrt{\frac{2m_0c^2}{m_p}} = \sqrt{\frac{2c^2}{1836}}$$
$$v = 9.9\times10^6\ m/s$$

A/c to De-Broglie's

$$\lambda = \frac{h}{m_p v} = 0.0004\ A^\circ \text{ Ans}$$

Q. Find the ratio of De-Broglie's Wavelength of an electron to that of a proton with the same energy

Find

$$\frac{\lambda_e}{\lambda_p} = ?$$

Date:

We know that,

$$\lambda = \frac{h}{p}$$
$$p = \frac{h}{\lambda}$$
$$m_e v = \frac{h}{\lambda_e}$$

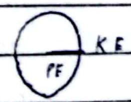
sq. both sides & × by 1/2

$$\frac{1}{2} m_e^2 v^2 = \frac{h^2}{2\lambda_e^2}$$
$$m_e\left(\frac{1}{2} m_e v^2\right) = \frac{h^2}{2\lambda_e^2}$$

∴ KE in terms of Broglie's λ

$$K.E_e = \frac{h^2}{2m_e\lambda_e^2}$$

$$E_e = \frac{h^2}{2m_e\lambda_e^2} \text{ —— (i)}$$



$E = KE + PE$

$E = KE + 0$

with the same energy

$$E_p = \frac{h^2}{2m_p\lambda_p^2} \text{ —— (ii)}$$
$$E_e = E_p$$

Equating (i) & (ii)

$$\frac{h^2}{2m_e\lambda_e^2} = \frac{h^2}{2m_p\lambda_p^2}$$

Date:

$$\frac{\lambda_e^2}{\lambda_p^2} = \frac{m_p}{m_e}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{1.67\times10^{-27}}{9.1\times10^{-31}}}$$

$$\frac{\lambda_e}{\lambda_p} = 42.81$$

Photo Electric Effect

E = work function + K.E

$$h\nu = \phi_0 + eV_0$$

$$\phi_0 = h\nu_0$$ threshold freq

Q. A substance is required for photocell operable the light of wavelength (~~40~~ 700nm - 400nm) → (1.77eV - 3.106eV) Which substance is suitable for the photocell

Titanium	$\phi_0 = 4.2eV$	✗
Tungsten	$\phi_0 = 4.5eV$	✗
~~Uranium~~		
Aluminium	$\phi_0 = 4.2eV$	
Barium	$\phi_0 = 2.5eV$	
Lithium	$\phi_0 = 2.3eV$	✓
Cesium	$\phi_0 = 1.9eV$	✓

$c = \nu\lambda$

Date:

$$\phi_0 = h\frac{c}{\lambda_0}$$

$$\lambda_0 = \frac{hc}{\phi_0}$$

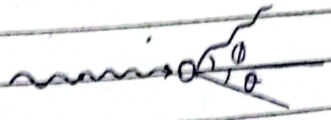
Q. When UV light of wavelength 4000Å strikes a cathode of photocell, a retarding potential of 0.4V is required to stop the ~~1~~ emission of electron. Calculate (i) light frequency (ii) Photon energy (E = hν) (iii) Work function (iv) Threshold frequency
** (v) net energy after the electron leaves the surface ($h\nu - \phi_0$)

(Solve at home)

Compton Effect

$$\Delta\lambda = \frac{h}{m_0c}(1-\cos\phi)$$

$$\lambda' - \lambda = \frac{h}{m_0c}(1-\cos\phi)$$



Paper Mark distribution Date:

20 - Ch01 - Theory
(5 Parts) 4 each Error to Radioactivity
(Vector) (BBR)

20 - Ch02 - Problems
(4 Parts) 5 each Vector to laser

20 - Ch03 - Problems
(4 Parts) 5 marks each Modern Physics
De-Broglie's to BBR

Date:

P.E.E

$$h\nu = \phi_0 + K.E$$

$$h\nu = h\nu_0 + eV_0$$

$$\boxed{\phi_0 = h\nu_0} \Rightarrow \frac{hc}{\lambda_0}$$

Q. Molebdymne has a work func. of 4.2eV.
i) Find cutoff wavelength and cut off frequency for photoelectric effect
ii) What is the stopping potential if the incident light has a wavelength of 180nm.

i) $4.20\ eV = 4.2 \times 1.6 \times 10^{-19} = 6.72 \times 10^{-19}\ J$

$$\nu_0 = \phi_0 / h = \frac{6.72 \times 10^{-19}}{h}$$

$$\nu_0 = 1.0142 \times 10^{15}\ Hz$$

$$\lambda_0 = \frac{hc}{\phi_0} = \frac{hc}{6.72 \times 10^{-19}} \text{ or } \frac{c}{\nu_0}$$

$$\lambda_0 = 2.958 \times 10^{-7}\ m$$

$$\lambda_0 = 2958\ A^\circ$$

ii) $KE = h\nu - \phi_0$

$$\cancel{k}\ eV_0 = h\nu - \phi_0$$

$$V_0 = \frac{h\nu}{e} - \frac{\phi_0}{e}$$

Date:

$$V_0 = \frac{hc}{\lambda e} - \frac{\phi_0}{e}$$

V_0

Q. Electrons are ejected from a metallic surface with a speed ranging upto $V = 4.6 \times 10^5$ m/s. When light with the wavelength of 625 nm is used

(i) What is the work func. of the surface

(ii) What is the cut-off freq. of the surface

$$KE = 1/2\, m V^2$$

$$KE = 1/2\, m_0 (4.6 \times 10^5)^2$$

$$KE = 9.64 \times 10^{-20} \text{ J}$$

$$\phi_0 = h\nu - KE$$

$$\phi_0 = \frac{hc}{\lambda} - KE$$

$$\phi_0 = 2.22 \times 10^{-19} \text{ J}$$

$$\phi_0 = h\nu_0$$

$$\nu_0 = \frac{2.22 \times 10^{-19}}{h}$$

Date:

Q. Lithium, Berrylium & Mercury has a work function of 2.30 eV, 3.9 eV and 4.5 eV respectively. When light with a wavelength of 400 nm is incident on each of these metals, find

(i) Which metal exhibit the photo-electric effect?

(ii) The max. KE for photon in each case: Lithium, Berrylium & Mercury

$$\phi_0 = \frac{hc}{\lambda_0}$$

$$\lambda_0 = \frac{hc}{\phi_0}$$

Lithium $\lambda_0 = 540$ nm.

$$KE_{max} = h\nu - \phi_0$$

$$KE_{max} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$KE_{max} = 0.8 \text{ eV}$$

Date:

Q. Calculate the De-Broglie's wavelength of an electron if the KE is 15 keV.

$$\lambda = \frac{h}{p}$$

$$mV = \frac{h}{\lambda}$$

{SBS.} $$(mV)^2 = \frac{h^2}{\lambda^2}$$

{$\times$ 1/2 BS} $$m\left(\frac{1}{2}mV^2\right) = \frac{1}{2}\frac{h^2}{\lambda^2}$$

$$m\,KE = \frac{h^2}{2\lambda^2}$$

$$\lambda^2 = \frac{h^2}{2mKE}$$

$$\lambda = \sqrt{\frac{h^2}{2mKE}}$$

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

$$\lambda = 1.002 \times 10^{-11}\ m$$

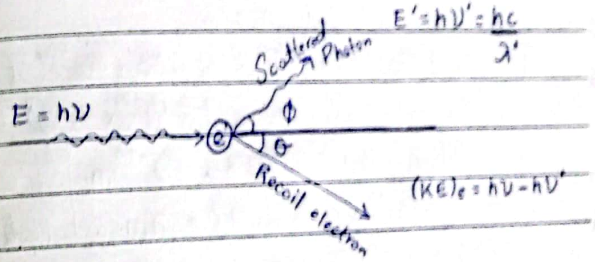
Alt $$\lambda = \frac{h}{\sqrt{2eV_0 m_0}}$$

Date:

Scattering

Coherent Scattering ⇓ Same phase ⇓ No change in wavelength

Incoherent Scattering ⇓ change in wavelength after collision ⇓ Compton Scattering



elastic collision → KE & momentum both conserved

$$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\phi)$$

change in wavelength

m_0 — mass of electron

ϕ — Angle of scattered photon

scattered incident Date:

$\Delta\lambda = \lambda' - \lambda$

∴ called Compton shift

Q. X-rays of wavelength 0.2000 00 nm are scattered from the block of the scattered x-rays are observed at an angle of 45° through the incident beam. Calculate the scattered wavelength

$$\lambda' - \lambda = \frac{h}{m_o c}(1 - \cos\phi)$$

$$\lambda' - 0.2000\,00 \times 10^{-9} = \frac{6.625 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^{8}}(1 - \cos 45)$$

$$\boxed{\begin{aligned} \lambda' &= 2.007 \times 10^{-10}\,\text{m} \\ \lambda' &= 0.2007\,\text{nm} \end{aligned}}$$

Ans

Ans

Q. Calculate the energy and momentum of photon of wavelength 700 nm.

$p = \frac{h}{\lambda}$

$$E = \frac{hc}{\lambda}$$

Date:

Q. X-rays having energy of 500 keV undergo compton scattering from a target the scattered rays are deflected at 37° relative to the incident way. Find the Compton Shift & ~~the~~ at this angle and the energy of recoiled electron.

$$E_{x\text{-ray}} = 300\,\text{KeV}$$
$$= 300 \times 10^{3}\,\text{eV}$$
$$E_{x\text{-ray}} = 4.8 \times 10^{-14}\,\text{J}$$

$\Delta\lambda = ??$ (Compton Shift)

$(KE)_e = h\nu - h\nu' = ??$

$$\Delta\lambda = \frac{h}{m_o C}(1 - \cos\phi)$$

$\phi = 37°$

$\Delta\lambda = 4.8 \times 10^{-13}\,\text{m}$ Ans

$$(KE)_e = h\nu - h\nu'$$
$$(K.E)_e = \frac{hc}{\lambda} - \frac{hc}{\lambda'}$$

for λ

$$E_{x\,ray} = \frac{hc}{\lambda}$$

Date:

$$\lambda = \frac{hc}{E_{x\text{-ray}}} = \frac{hc}{4.8 \times 10^{-14}}$$

$$\lambda = 4.14 \times 10^{-12} \text{ m}$$

<u>for λ'</u>

$$\Delta\lambda = \lambda' - \lambda$$

$$\lambda' = \Delta\lambda + \lambda$$

$$\lambda' = 4.6 \times 10^{-12} \text{ m}$$

$$(KE)_e = h\nu - h\nu'$$

Q. 0.110 nm photon ~~is the~~ is incident on a stationary electron. After the collision electron move forward & photon recoiled back.
i) Calculate momentum & KE of electron

Date:

$$KE = \frac{1}{2} mV^2$$

$$KE = \frac{P^2}{2m}$$

(i) P of e^- = ??

(ii) $(KE)_e = h\nu - h\nu'$

$$= \frac{hc}{\lambda} - \frac{hc}{\lambda'}$$

$$\lambda' = \frac{h}{m_0 c}(1 - \cos\phi) + \lambda$$

$\therefore \phi =$

$$E = h\nu$$

$$E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E}$$

$$\lambda' = 0.115 \text{ nm}$$

$$\lambda = 0.110 \text{ nm}$$

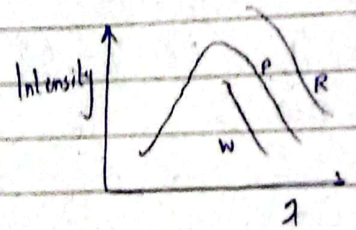
$$(KE)_e = 1.8 \times 10^{-17} \text{ J}$$

$$P = 4.2 \times 10^{-24}$$

Ans

Date:

The study of black body radiation is important because it gives & explains the nature of radiation. The cavity radiator has a small whole which does not allow any radiation to escape from the cavity once it has entered inside it. Therefore, a cavity radiator may behave as a perfect absorber so it is called a black body



Consider the curve of a black body at T = 1646K.
The curve explains the distribution of energy for different wavelengths.
Different scientists have explained the curve which are as follows:
There are 3 classical laws of BBR
Wein's Law: This theory explains only the distribution of energy in short wavelength but does not explain the energy distribution in long wavelength
Mathematically,

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$\lambda T = const = 0.0029$

Rayleigh Jean's Theory: This theory ~~explains~~ agrees on long λ but does not explain the energy distribution in short λ.
Mathematically,

$$I(T) \propto T^4$$

$$I(T) = \sigma T^4$$

$$\sigma = 5.67 \times 10^{8} \text{ Watt}/m^2 K^4$$

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$\lambda_{max} = 0.0028 / T$

$I(T) = \sigma T^4$

$E = h\nu$

Energy exchanges takes place in discrete bundles or 'quanta'. The higher the frequency the more the energy bundles has to be emitted.

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Q. The radius of the Sun is 6.96×10^8 m and its total power output is 3.77×10^{26} W. Assuming the sun's surface emits a black body radiation, calculate its surface's temperature?

$R = 6.96 \times 10^8$ m

$P = 3.77 \times 10^{26}$ W

$T = ??$

$\frac{P}{A} = \sigma T^4$

$A = 4\pi R^2$

$A = 6.09 \times 10^{18}$ m

$\frac{3.77 \times 10^{26}}{6.96 \times 10^8} = 6.67 \times 10^{-8} T^4$

$T =$

Using the answer of part a, find

b) $\lambda_m \times T = 2.8 \times 10^{-3}$

λ

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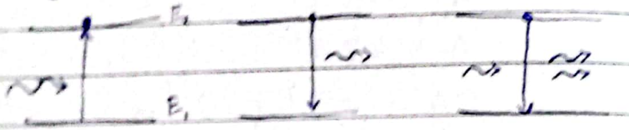
LASER Theory

-> Laser is basically a light source, the radiation that it emits is not fundamentally different ~~from~~ than any other form of radiation

-> It is highly directional, is of high brightness, and its amplitude remains constant

Types of Emission:



① Absorption: occurs when the photon energy is equal to the difference of energys b/w two levels $(E_2 - E_1)$, thus causing excitation

② Spontaneous Emission: When an atom in excited state E_2 goes to the ground state E_1, by spontaneously emitting of photon of frequency $\nu = \frac{E_2 - E_1}{h}$. This process is known as Spontaneous Emission.

In such case.

-> The emitted photon can move in random direction

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-> Photons ~~fro~~ emitted from various atoms in the assembly have no phase-relationship thus the radiation given out by such source is incoherent

③ Stimulated Emission: When an atom in excited state E_2 interacts with an incident photon of right frequency and thereby induce to move to the ground state E_1, with the difference of energies as a photon of same energy released. ~~&~~ This process is known as Stimulated Emission.

In such case:

-> emitted photon travels in same direction in which incident photon is moving

-> These two photons are in phase